

Integrating societal preferences: Decentralization of the European energy network

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Abstract

Filler

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PyPSA-EUR, Energy systems modeling, Fairness in energy transitions

1. Introduction

In a world where the necessity of reducing the carbon emissions of our energy networks is no longer a question but a requirement careful and deliberate planning of the energy system of our future is needed. When the Paris Climate agreement was signed into action in 2015 it signaled the commitment of the United Nations to reducing their overall carbon footprint [1]. This common goal and understanding of the necessity of decarbonization was a crucial step towards a more sustainable world, and providing a common base understanding of what needs to be done, but also provides a massively complicated technological challenge that must be solved in order for the agreement to be fulfilled.

Somewhat unique to the issue of decarbonization in the European Union, is the fact that, although the Paris Climate Agreement signifies the common goal of decarbonization, The Union is still comprised of many different countries with many different political ideologies, laws, societal preferences and goals of their own. This means that although it has long been known that a shared and interconnected energy grid is the most efficient and economically preferable solution, as seen for example in [2], reality becomes a lot more complicated as the member nations also must have their own best interest in mind.

Historically the impact of the union being made up of self-centered actors has often been discarded when trying to construct the electricity grid of the future, if for no other reason than the fact that implementing social preferences and societal paradigms in strict mathematical models of the European energy system is a challenging problem [3]. These models that typically aim to reduce the overall cost of the system as the primary objective are often meant to be used as the basis for developing the future energy network and legislation. However using this least-cost optimization approach without considering social paradigms in the European Union tends to produce system models that, although being economically efficient at achieving decarbonization targets, that are intrinsically tied to the assumption that the EU has complete control over the member nations individual energy grids.

This approach of reducing the optimization problems to purely techno-economic problems that prioritize cost reduction over feasibility has helped achieve significant insights into decarbonization strategies for the European Union [4], but tends to lead to the energy system becoming heavily centralized [5]. Cen-

tralization in the context of energy systems refer to the majority of resources being pooled into a few areas that are favorable to the system in some way, be that installing a lot of wind turbines in a region with a lot of wind, maintaining energy production in regions close to where the energy is needed or any number of other mechanisms that result in a proportionally large investment into infrastructure in a single region. While centralization as a strategy is often the most cost efficient strategy it leads to a suite of other challenges.

Two such challenges are those of physical security and and political feasibility. With the recent surge in so-called hybrid attacks on European energy infrastructure[6], the focus on physical security has increased. If a system is overly centralized it means that a large amount of critical infrastructure is installed on a relatively small area. This makes the infrastructure an obvious target for such hybrid attacks as a single hostile action can impact more infrastructure and more people. As for the political feasibility, nations who end up as net energy importers become less self sufficient and increasingly reliant on the net exporting regions. This shift in power-balance heavily favors the net exporters and as such may face resistance from net importing nations that wish to remain at a high degree of self sufficiency.

The objective of this masters thesis is to incorporate societal paradigms and priorities in the form of decentralization into a macro-scale energy network model of the EU, and to analyze and expose the mechanisms behind decarbonization and its interactions with decentralization. The main hypothesis is that, by incorporating decentralization, the resulting networks will illuminate alternative transition pathways. This in tun may then help develop a workflow that by incorporating societal paradigms and reviewing their system impacts, will help close the gap between the models used to inform energy policy makers and the results of the proposed policies. Although this workflow and methodology may in time prove relevant for policy makers to understand, the target audience of this master’s thesis remains peers in the field of energy system modeling who may draw inspiration from the intersection between societal paradigms and strict mathematic modeling that this work represents. It is also for this reason that all the data and code used to produce the results which can be seen in this paper is openly available in a public repository under an open science license[7].

2. Nomenclature

Table Example

C1 R1	C1 R2
C2 R1	C2 R2

3. Methodology

Many energy modeling frameworks and pieces of commercially available software for optimizing nodal-networks exist. The work represented by this paper utilizes the open-source framework PyPSA (Python for power system analysis [8]). The reasoning for this is twofold: Firstly PyPSA is completely free and open source to use. This encourages open and collaborative science as well as transparency in the philosophy behind the energy system modeling. Secondly PyPSA consists of two separate parts. A modeling workflow based in the Python programming language, and a large database containing relevant open source data for constructing models, particularly of the European Union. This dataset is referred to as PyPSA-Eur[9] and is invaluable in its quality of data for building macro energy system models in the PyPSA workflow. This masters thesis utilizes both parts of the PyPSA environment. Acting as a baseline for all of the network topology for the European Union is a pre-compiled network developed with PyPSA-EUR using the default settings as published in the Zenodo repository [10].

The simple philosophy behind the PyPSA models is that a network is perceived as multiple interconnected nodes. These nodes and connections are then given a set of constraints that define the feasible solution space. Examples of constraints could be that each node is assigned an electrical load that must be serviced based on historical consumption, capacity factor time-series based on solar- and wind-patterns, or maximum or minimum capacity constraints like a maximum allowable power flow from one node to another representing the transmission infrastructure in the network. The network used in this master's thesis is a 37 node network representing the 25 (excluding the island states of Malta and Cyprus) member nations of the EU. Additionally Switzerland, Norway, Bosnia-Herzegovina, Albania, Montenegro, North Macedonia, The United Kingdom and Serbia are included as their electrical infrastructure is connected to that of the EU. The following nations are also represented by more than one node as they have large populations on islands: Denmark, Spain, Great Britain and Italy.

These nodes are then interconnected with HVDC and AC links and lines respectively. The nodes are then each assigned load patterns, and

3.1. Optimization strategy

Optimizing complex energy networks with many variables and constraints is computationally very intensive. For this reason the optimizations performed for this master's thesis have exclusively been performed on a High Performance Computing (HPC) environment. More specifically the UCloud HPC operated by SDU [11]. The optimizations were typically run on a set of 4 machines, each with an Intel Xeon Gold 6130 CPU and took on average 12 core-hours apiece to achieve an optimal solution. The optimization algorithm was chosen as the Barrier method in the commercial optimizer by Gurobi, chosen primarily for its numerical stability [12].

3.1.1. Objective functions and constraints

Objective functions in PyPSA and interacting with Gurobi. Adding the de-carbonization constraint to the least cost system

In order to optimize the network first an objective function must be defined. An objective function is, as the name suggests, the function which when minimized or maximized leads to the objective of the optimization to be achieved. The objective function for finding the cheapest possible electrical system is defined as minimizing the sum of all the costs:

Equation:

$$\min \left(\sum_n \text{generation costs} + \text{storage costs} + \text{transmission costs} + \sum_{n,t} \text{variable costs} \right) \quad (1)$$

Here n denotes the set of all nodes, and t denotes the set of all snapshots during the timeframe. However in order to construct a feasible solution space the objective function must be defined by a set of bounding constraints. Initially an equality constraint is defined so that, for each snapshot, all nodes must have their electricity demand met by a combination of energy production and nodal balancing:

$$\text{generation}_{n,t} + \text{balance}_{n,t} = \text{demand}_{n,t} \quad \longleftrightarrow \quad \lambda_{n,t} \quad \forall n, t \quad (2)$$

3.2. Least-cost system

Introduction to least-cost systems and their purpose as a baseline.

3.2.1. Modeling

Modeling assumptions and methodology. Describing network topology and cost modeling.

3.3. Decentralized systems

Introduction to the secondary decentralizing sweeps

3.3.1. Measures of decentralization

Discussion on Standard deviation approach vs Gini Approach

3.3.2. Linear systems

Talking points on avoiding non-convex solution spaces

3.3.3. Added constraints

Discussing the K-constraints introduced for decentralization

3.3.4. Avoiding constraint interactions

Talking about how the normal K approach does not work as it leads to over de-carbonization

4. Results

Introduction to results

4.1. Least-cost systems

Maps and figures showing a few least-cost networks, their optimized topology and key figures like system cost and marginal prices.

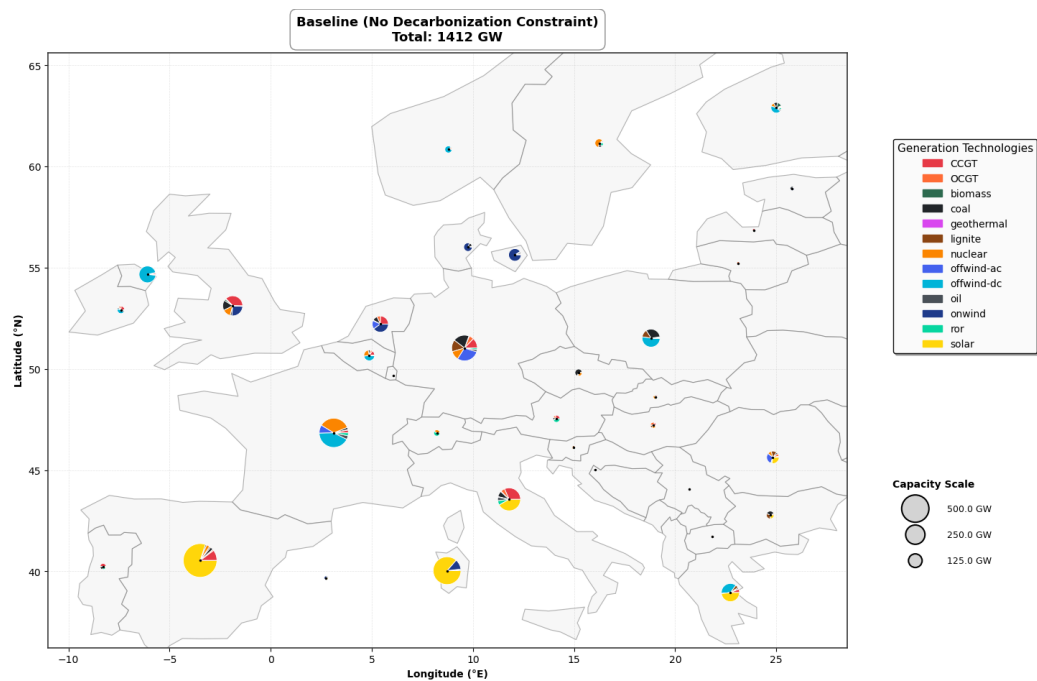


Figure 1: Figure text

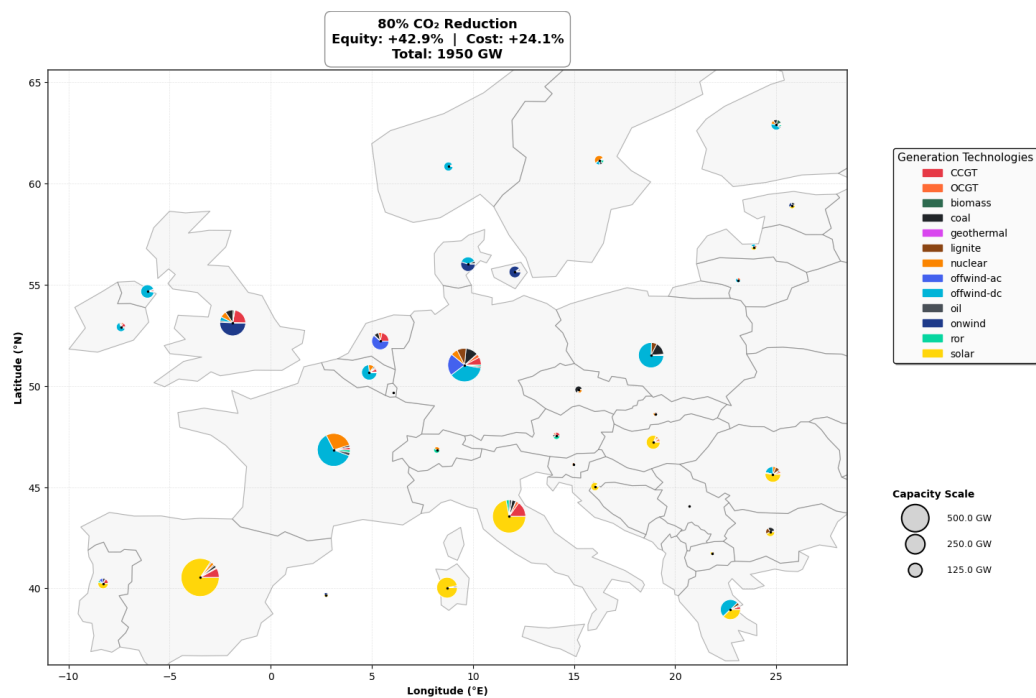


Figure 2: Figure text

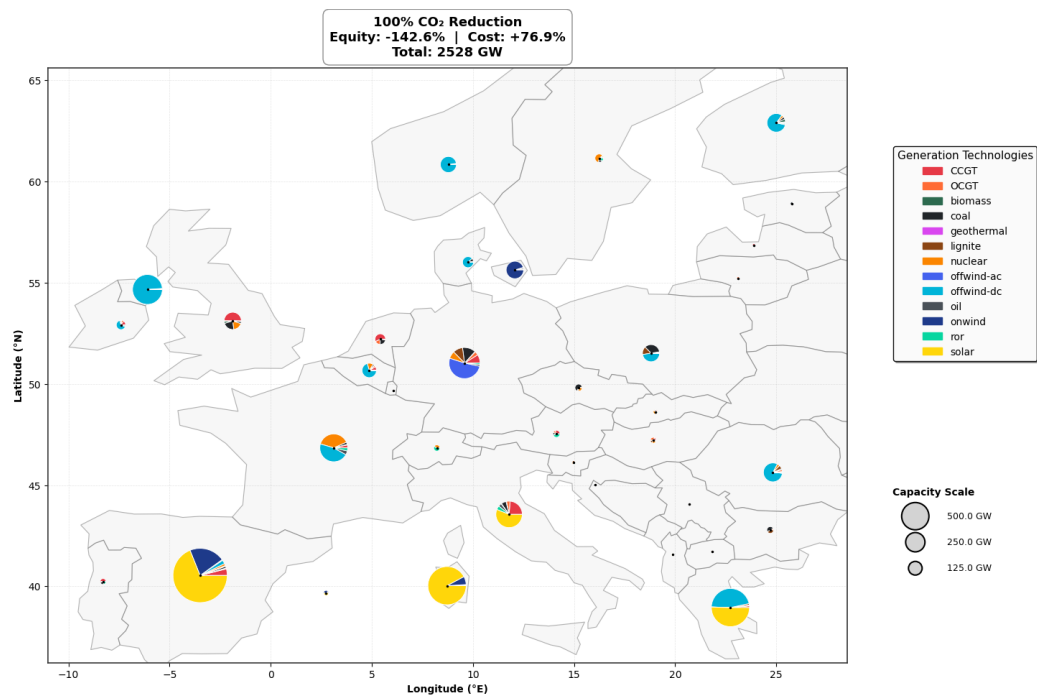


Figure 3: Figure text

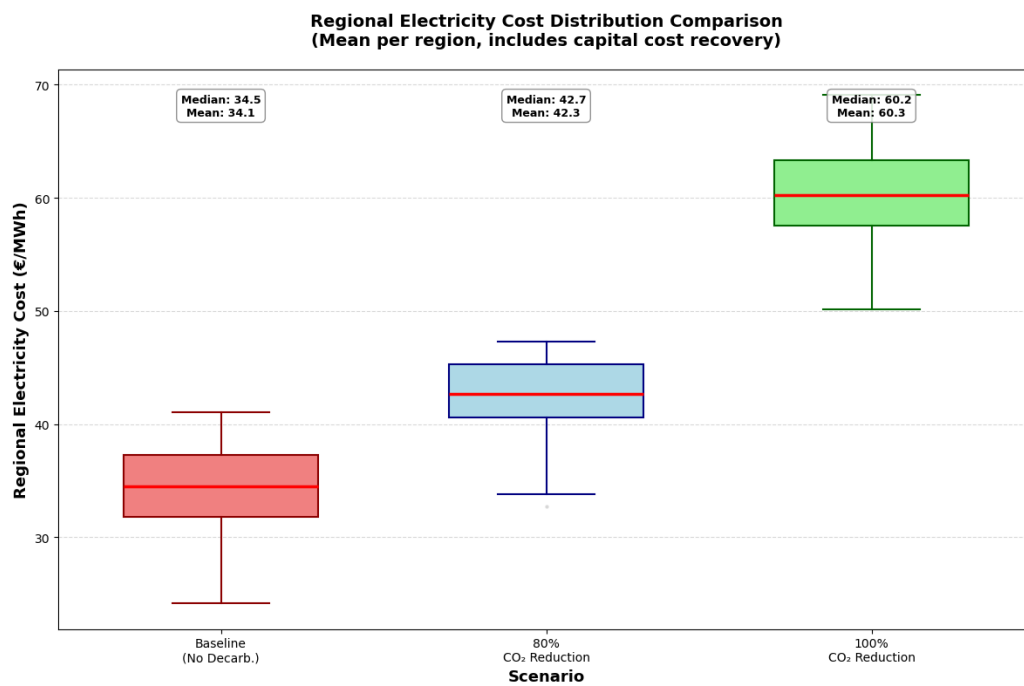


Figure 4: Figure text

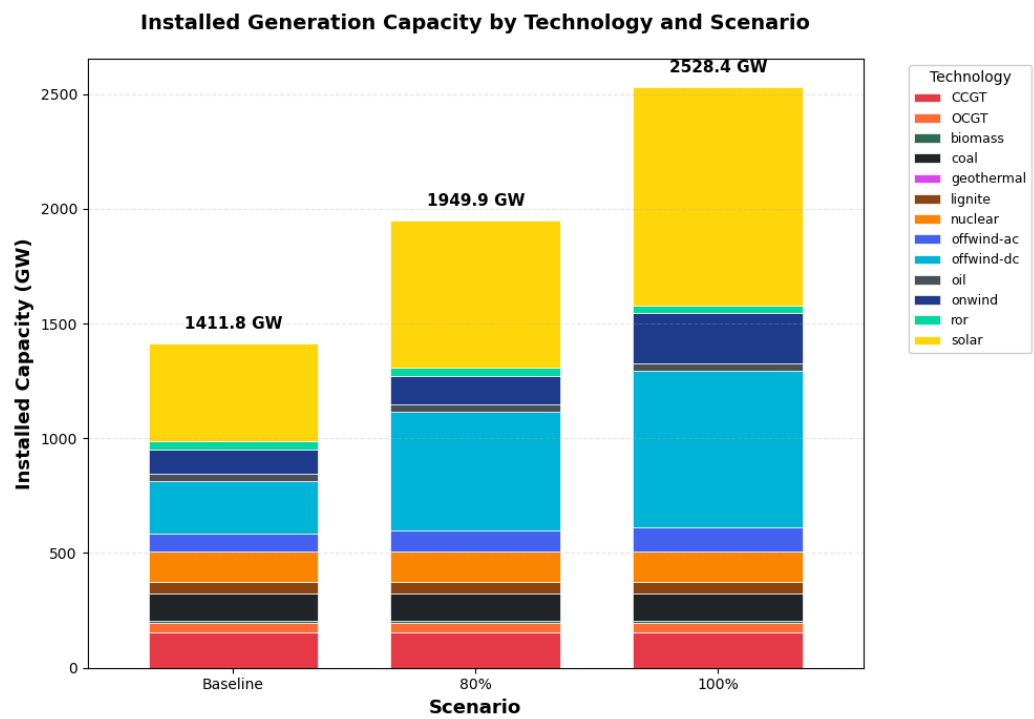


Figure 5: Figure text

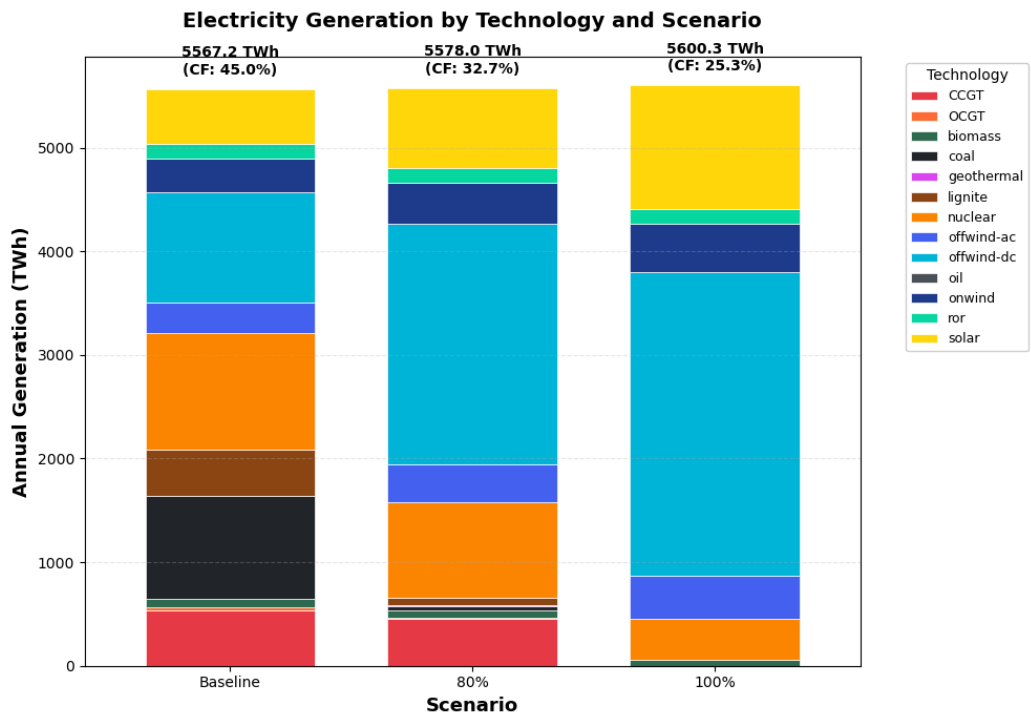


Figure 6: Figure text

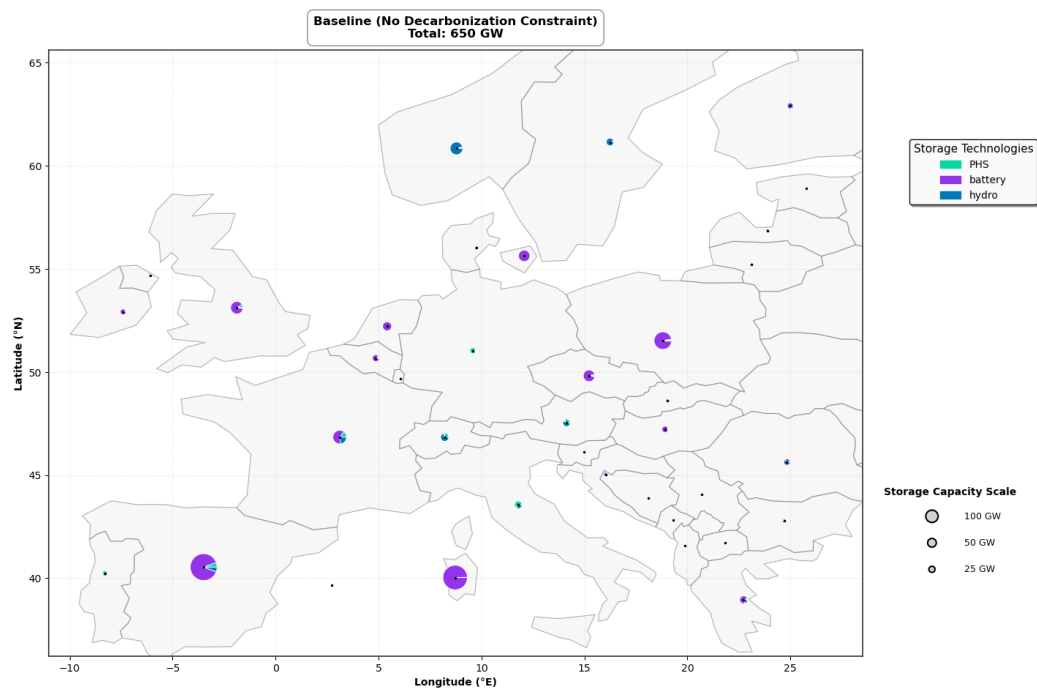


Figure 7: Figure text

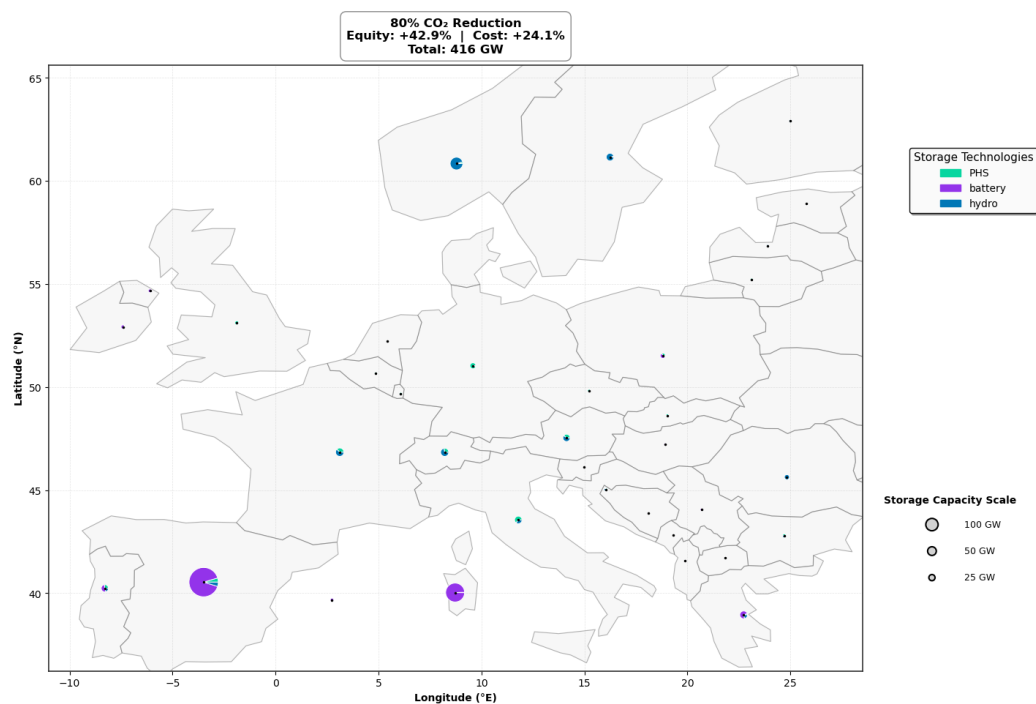


Figure 8: Figure text

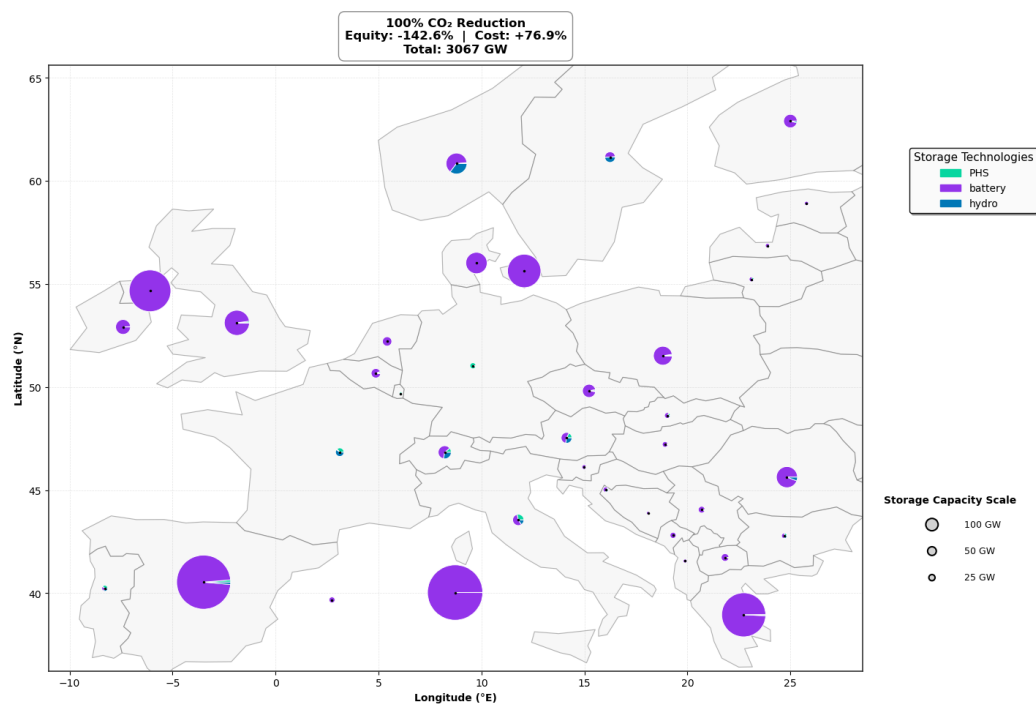


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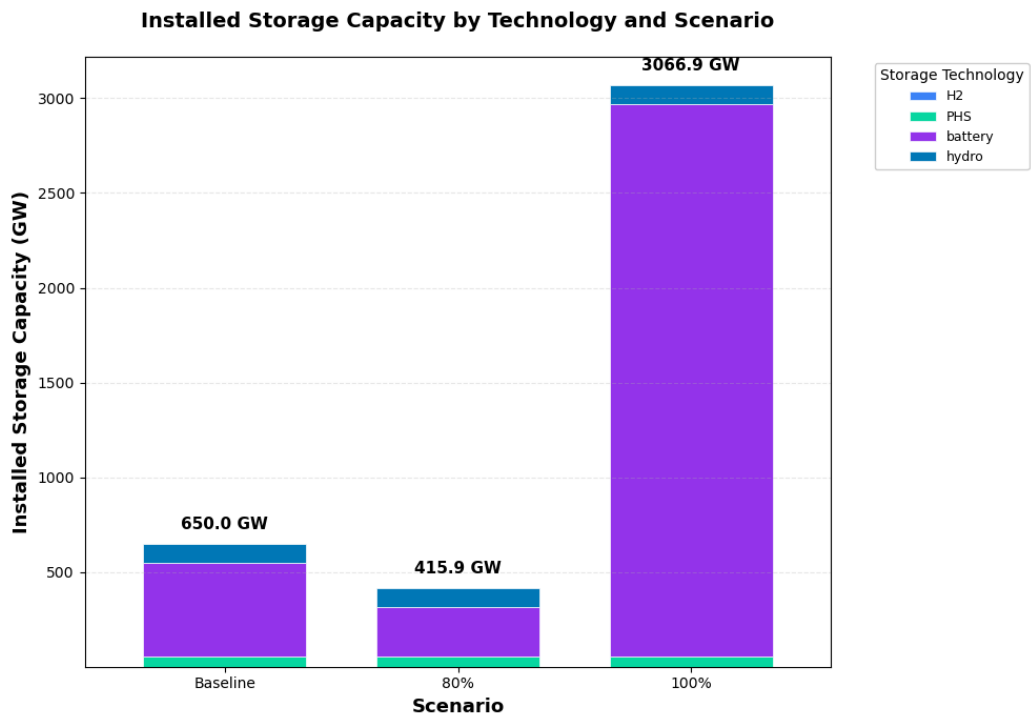


Figure 10: Figure text

4.1.1. Sanity checks and validations

Comparing systems to real systems

4.1.2. Decarbonization trends

Analyzing trends in key figures and plots for the decarbonizing systems

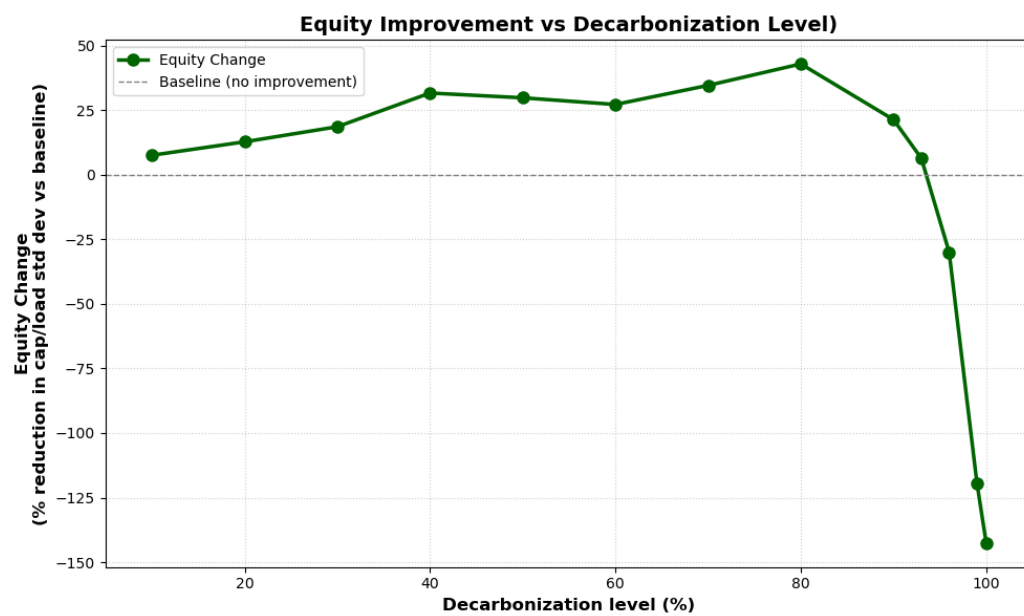


Figure 11: Figure text

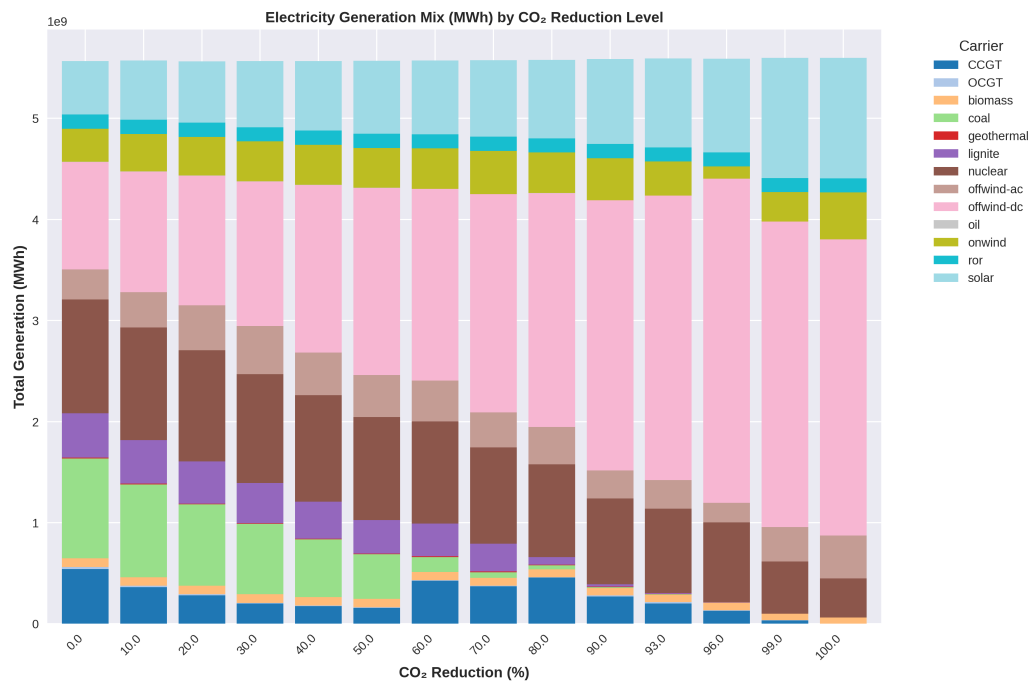


Figure 12: Figure text

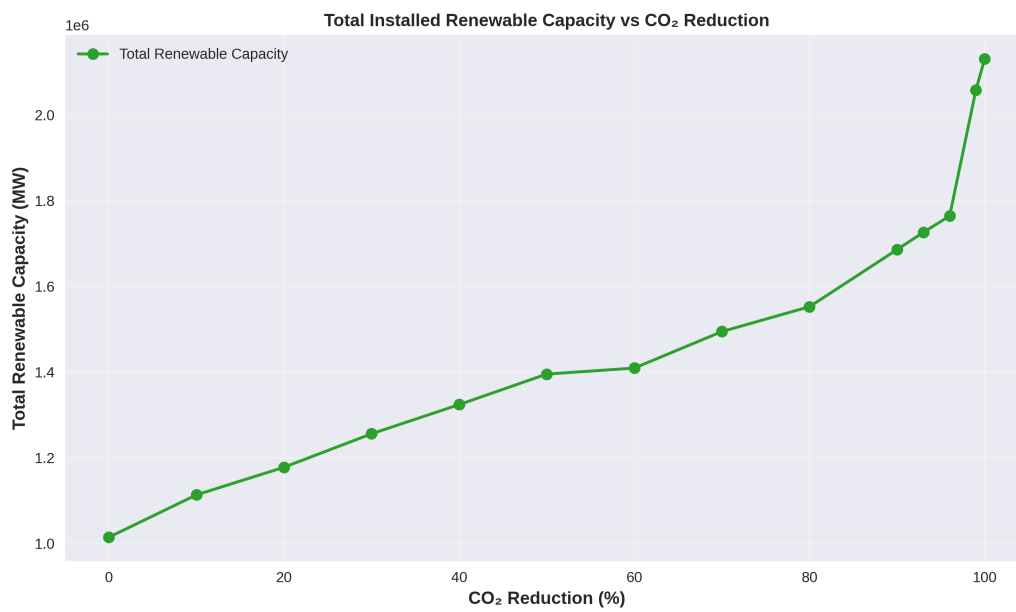


Figure 13: Figure text

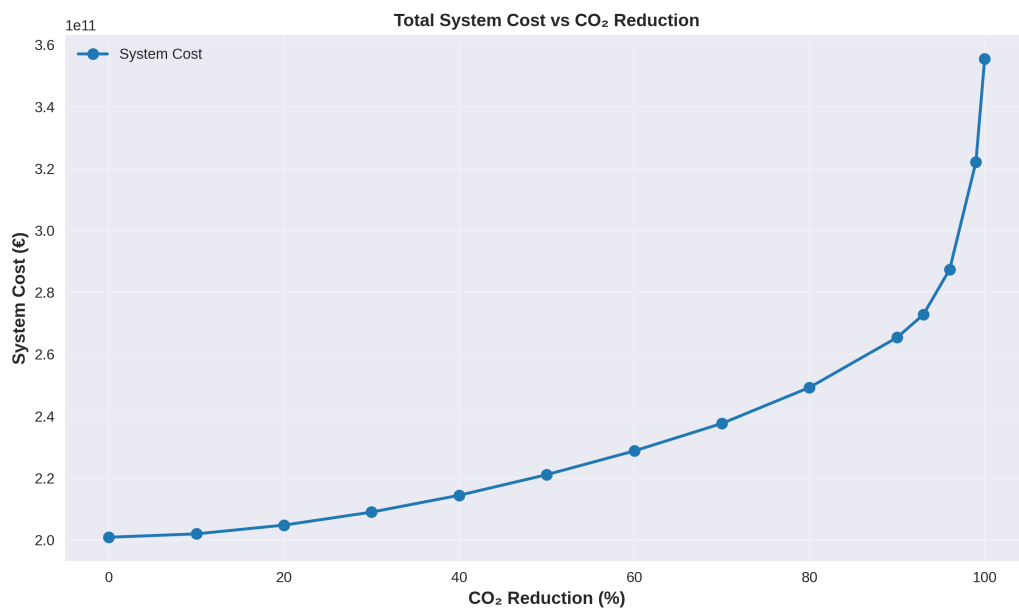


Figure 14: Figure text

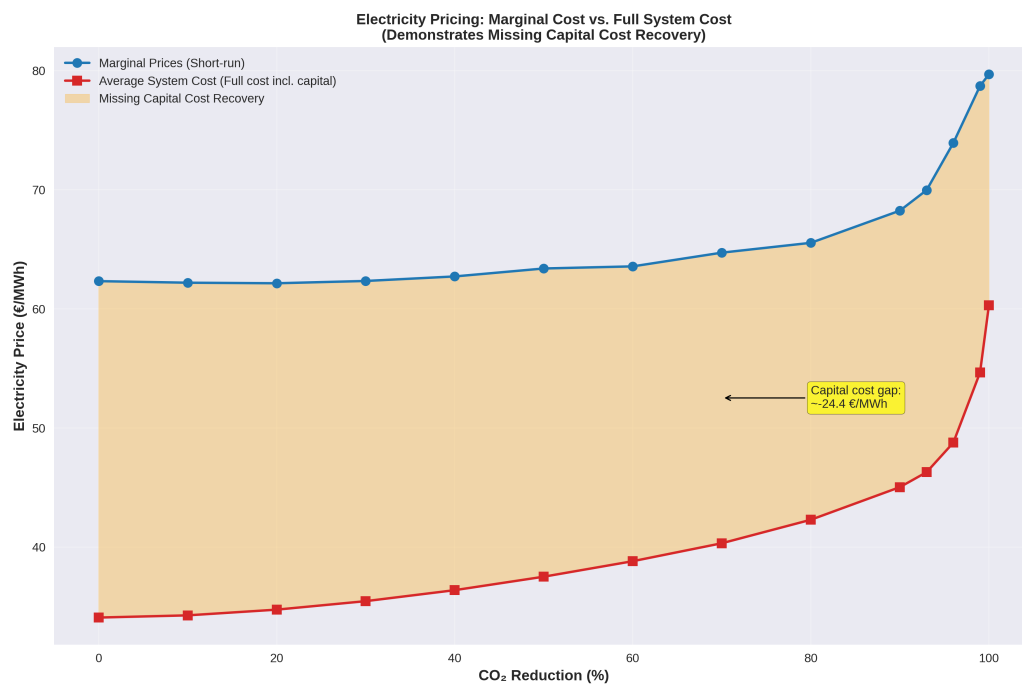


Figure 15: Figure text

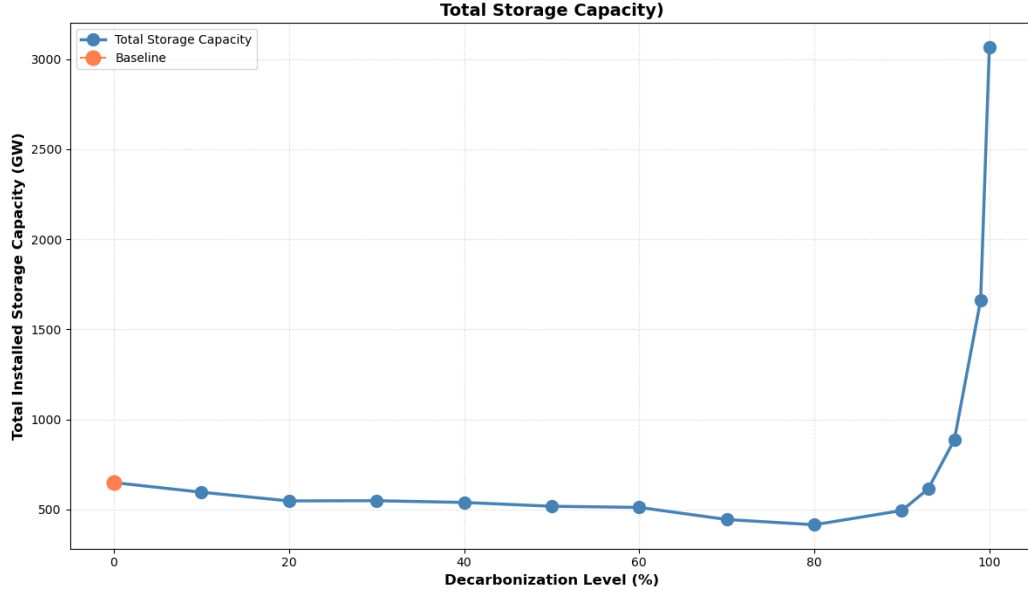


Figure 16: Figure text

4.1.3. *Driving Mechanisms in decarbonizing systems*

Discussing driving mechanisms behind trends like load-following and resource optimization

4.2. *Decentralizing systems*

Introduction to the decentralizing systems showing a few networks and comparing their topology in contrast to their baselines.

4.2.1. *Sanity checks and validations*

Comparing the key figures like marginal prices to those from real networks

4.2.2. *Decentralization trends*

Analyzing how decentralization, de-carbonization and key figures interact. Plotting 3-d response surfaces and discussing shape of solution spaces.

4.2.3. *Driving Mechanisms in decentralizing systems*

Talking about mechanisms that produce the interactions seen above

5. Discussion

Introduction to discussion

5.1. *Economic trends and societal paradigms*

What can interactions can be taken from my results and how does this align with expectations and real-world trends

5.2. *Future policy implications*

Discussing how this workflow can be used to analyze and incorporate paradigms to review implications of energy policy.

5.3. *Uses for methodology*

How else could this approach be used and how would this expand the workflows use-case.

6. Conclusion

7. Assumptions

8. Acknowledgments

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10. Appendices

Cross reference: (1)

Inline equation: CO_2

Figure reference and figure: 1:

List:

- S represents the set of snapshots,
- G_r is the set of generators located in region r ,
- $p_{g,s}$ is the power output of generator g at snapshot s ,
- e_g is the emission factor of generator g , and
- w_s is the weighting factor for snapshot s .